Car Price Prediction

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
car=pd.read_csv('quikr_car.csv')
car.head()
```

```
car.shape


car.info()
```

some changes

- names are inconsistent
- names have company names attached to it
- some names are spam like 'Maruti Ertiga showroom condition with' and 'Well mentained Tata Sumo'
- company: many of the names are not of any company like 'Used', 'URJENT', and so on.
- year has many non-year values
- year is in object. Change to integer
- Price has Ask for Price
- Price has commas in its prices and is in object
- kms_driven has object values with kms at last.
- It has nan values and two rows have 'Petrol' in them
- fuel_type has nan values

Cleaning Data

year has many non-year values

```
car['year'].unique()
car['Price'].unique()
car['kms_driven'].unique()
car['fuel_type'].unique()
```

cleaning the data

```
backup=car.copy()
```

year has many non-year values

```
car['year'].str.isnumeric()

car=car[car['year'].str.isnumeric()]
```

year is in object. Change to integer

```
car['year'].astype(int)

car['year']=car['year'].astype(int)
```

```
car.info()
```

Price has Ask for Price

Price has commas in its prices and is in object

```
car=car[car['Price']!='Ask For Price']
car['Price']=car['Price'].str.replace(",","").astype(int)
```

```
car.info()
```

kms_driven has object values with kms at last.

```
car['kms_driven']

car['kms_driven']=car['kms_driven'].str.split().str.get(0).str.replace(',','')
```

It has nan values and two rows have 'Petrol' in them

```
car=car[car['kms_driven'].str.isnumeric()]
```

```
car['kms_driven']=car['kms_driven'].astype(int)
```

fuel_type has nan values

```
car.info()

car=car[~car['fuel_type'].isna()]
```

```
car.shape
```

name and company had spammed data...but with the previous cleaning, those rows got removed.

```
car['name'].str.split().str.slice(0,3)
```

```
car['name'] =car['name'].str.split().str.slice(0,3).str.join(' ')
```

Resetting the index of the final cleaned data

```
car=car.reset_index(drop=True)
```

```
car.info()

car.describe()
```

there is one value whose price has outlier we will keep below 60L price

```
car=car[car['Price']<6e6].reset_index(drop=True)
```

now save the cleaned data file

```
car.to_csv('Cleaned_Car_data.csv')
```

now distribute the x & y coordinate

```
X=car[['name','company','year','kms_driven','fuel_type']]
y=car['Price']
```

now import the train test split file and necessory library

```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=34)
```

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import make_column_transformer
from sklearn.pipeline import make_pipeline
from sklearn.metrics import r2_score
```

```
ohe=OneHotEncoder()
ohe.fit(X[['name','company','fuel_type']])
```

```
column_trans=make_column_transformer((OneHotEncoder(categories=ohe.categories_),[
'name','company','fuel_type']),remainder='passthrough')
```

```
lr=LinearRegression()
```

```
pipe=make_pipeline(column_trans,lr)
```

```
pipe.fit(X_train,y_train)
```

```
y_pred=pipe.predict(X_test)
```

```
r2_score(y_test,y_pred)
```